Letter Frequency Attack on Monoalphabetic Substitution Cipher

37. Write a C program that can perform a letter frequency attack on any monoalphabetic substitution cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify “give me the top 10 possible plaintexts.”

Aim:

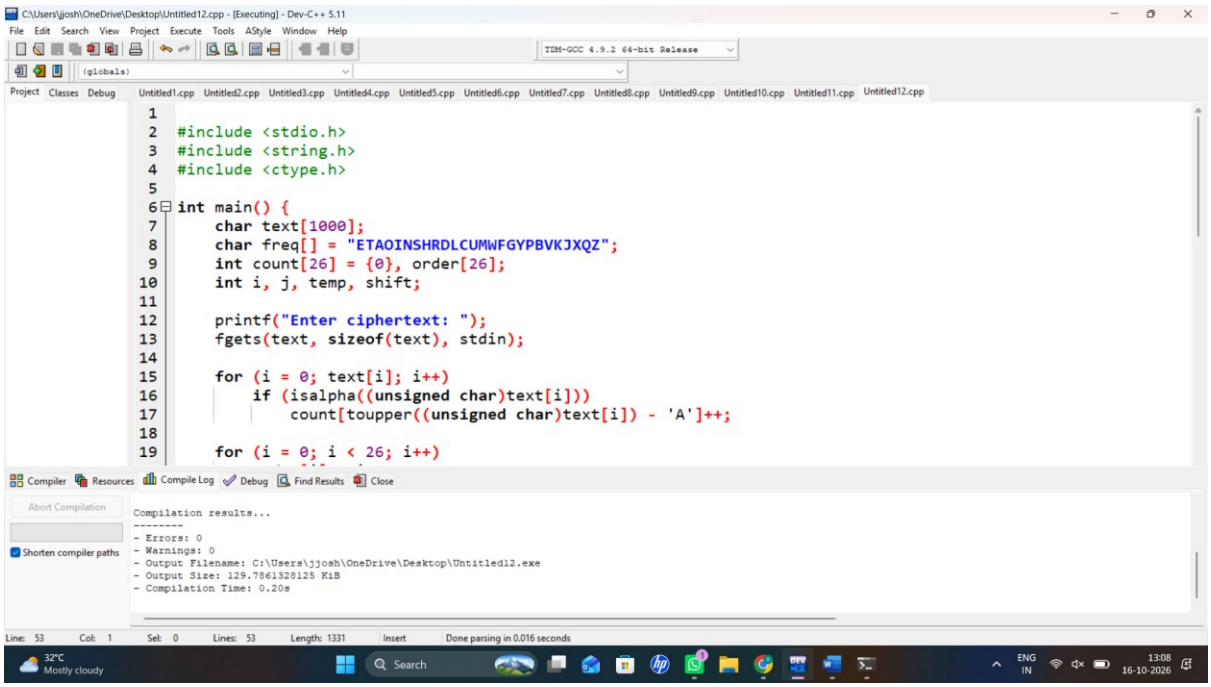
To implement a frequency-analysis attack in C that estimates plaintext from a monoalphabetic substitution ciphertext without manual letter-by-letter decoding.

Algorithm:

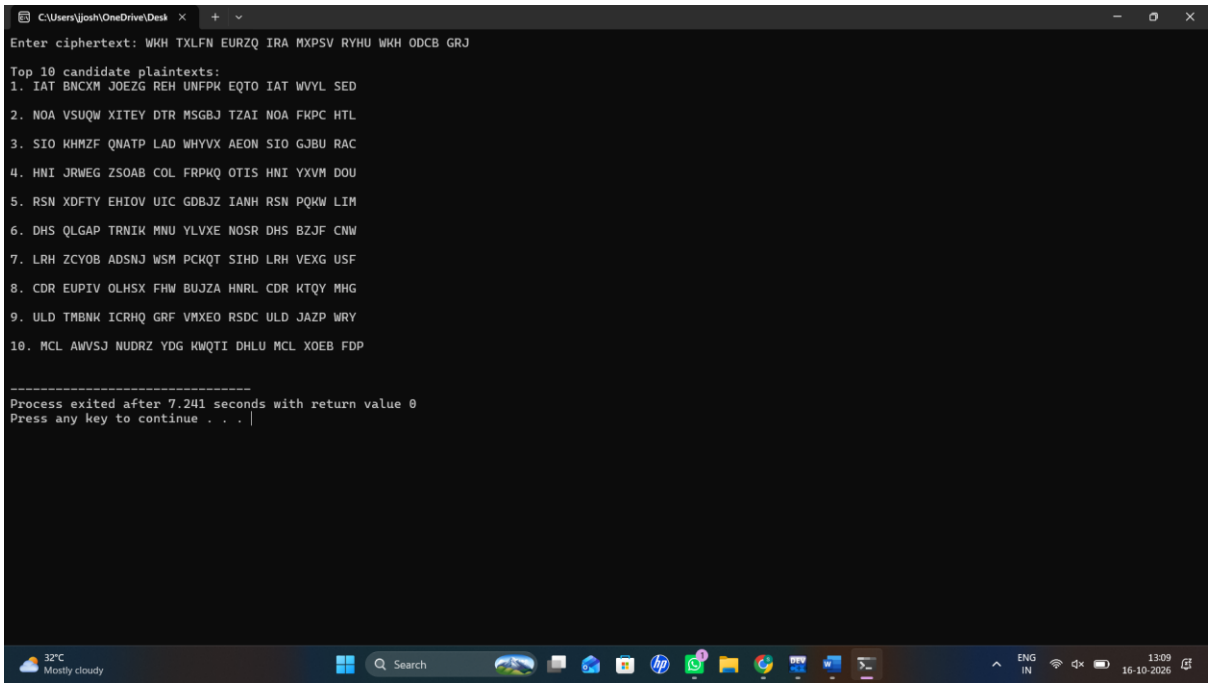
1. Read the ciphertext.
2. Count the occurrences of each alphabetic character.
3. Sort the letters by decreasing frequency.
4. Match the most frequent ciphertext letters with common English letters.
5. Generate possible plaintext candidates using different frequency alignments.
6. Display the candidate plaintexts in estimated order of likelihood.

Program:

This beginner-friendly version uses English letter frequencies and produces ten candidate decryptions. It is a heuristic, so the correct plaintext is not guaranteed to appear first.



Input/Output:



The exact candidates are generated when the program runs. This is a simplified frequency-ranking demonstration; full substitution-cipher cracking typically requires dictionary scoring, n-gram statistics, or optimization over possible substitution keys.

Result:

A frequency-analysis program was implemented to generate candidate plaintexts automatically. The method demonstrates how statistical properties of English can help attack monoalphabetic substitution ciphers.